

NIDS Technical Architecture Specification

System Topography, Component Relationships, and Pipeline Flow | Generated: 2026-07-28

1. Configuration & Data Ingestion

The configuration layer uses Pydantic schema validation for thread-safe settings loading. Data loaders ingest PCAPs, NetFlow, and CSV telemetry.

2. Inference Engine & Alert Subsystem

The inference pipeline processes 5-tuple flows in real time, scoring risk levels (Low, Medium, High, Critical) and persisting alert events into SQLite and SIEM exporters.

3. Security & Governance

JWT Bearer authentication, PBKDF2 password hashing, and 5-Tier RBAC protect administrative endpoints and SOAR playbooks.